



# Major Features of the 2021 WHO Classification of CNS Tumors

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Accepted: 6 May 2022 / Published online: 16 May 2022  
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## Abstract

Advances in the understanding of the molecular biology of central nervous system (CNS) tumors prompted a new World Health Organization (WHO) classification scheme in 2021, only 5 years after the prior iteration. The 2016 version was the first to include specific molecular alterations in the diagnoses of a few tumors, but the 2021 system greatly expanded this approach, with over 40 tumor types and subtypes now being defined by their key molecular features. Many tumors have also been reconceptualized into new “supercategories,” including adult-type diffuse gliomas, pediatric-type diffuse low- and high-grade gliomas, and circumscribed astrocytic gliomas. Some entirely new tumors are in this scheme, particularly pediatric tumors. Naturally, these changes will impact how CNS tumor patients are diagnosed and treated, including clinical trial enrollment. This review addresses the most clinically relevant changes in the 2021 WHO book, including diffuse and circumscribed gliomas, ependymomas, embryonal tumors, and meningiomas.

**Keywords** WHO · Glioma · Astrocytoma · Ependymoma · Embryonal · Meningioma

## Introduction

The fifth edition of the WHO Classification of Central Nervous System Tumors was released at the end of 2021, a mere 5 years after the fourth edition was published [1, 2]. Novel techniques such as next generation sequencing, RNA expression profiling, and DNA methylation profiling have paved the way for the discovery and classification of new entities, as well as more precise classification and stratification of existing tumors [3]. These rapid changes in the understanding of the molecular features that define CNS tumors

fostered the Consortium to Inform Molecular and Practical Approaches to CNS Tumor Taxonomy (cIMPACT-NOW) in 2017 to quickly provide updates in the pathological workup of CNS tumors between WHO editions [4]. The fifth edition of the WHO Classification of Central Nervous System Tumors incorporates this updated understanding of the molecular underpinnings of CNS tumors while maintaining their histopathologic roots. The purpose of this review is to discuss how these updates will impact clinical care, focusing on adult-type diffuse gliomas, pediatric-type diffuse gliomas, circumscribed astrocytic gliomas, ependymal tumors, and embryonal tumors (summarized in Table 1), as these are the entities with the most dramatic changes.

Before discussing specific tumors, it is worth mentioning a few general changes in grading. The first is that WHO grades, which were previously listed in Roman numerals, are now listed in Arabic numerals. In addition, grading is now done within tumor types as part of the integrated diagnosis, so although grades still correspond to natural history, there is not necessarily perfect equivalence between the same numerical grade in different types of tumors, i.e., a grade 4 medulloblastoma does not necessarily mean the same prognosis as a grade 4 IDH<sup>wt</sup> glioblastoma. Also, the term “anaplasia” is no longer employed, instead only “WHO grade 3” is used in the diagnosis. Finally, since a grade 2 glioma (for example) does not necessarily have the same general

Invited review, “Special Issue on Therapeutic Advances in Neuro-oncology”.

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**Table 1** Glial, ependymal, and embryonal tumors in the 2021 WHO Classification System

|                                          | Tumor type                                                             | Definition                                                                                                                                                                                     | Recurrent genetic abnormalities                                                                                                                                      | Grade                                                                                       | Management and prognosis                                                                                                                        |
|------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Adult-type diffuse astrocytoma</b>    | Glioblastoma, IDH-wildtype                                             | Diffuse, high-grade astrocytoma lacking IDH and histone mutations that has concomitant + 7/-10, EGFR amplification, TERT promoter mutations, mitoses, necrosis, or microvascular proliferation | Chromosomes + 7/-10, EGFR amplification, TERT promoter mutations, and frequently MGMT promoter methylation or mutations in PI3K pathway, p53 pathway, CDK4/6 pathway | 4                                                                                           | Resection and radiation ± adjuvant alkylating therapy are current standard. MGMT promoter methylation associated with better treatment response |
|                                          | Diffuse astrocytoma, IDH-mutant                                        | Diffuse astrocytoma with activating mutation in IDH1 or IDH2                                                                                                                                   | IDH1/2, ATRX, TP53, CDKN2A                                                                                                                                           | 2–4, depending on necrosis, mitoses, microvascular proliferation, and CDKN2A deletion       | Resection and radiation ± alkylating therapy. More indolent than glioblastoma, IDH-wildtype, but CDKN2A deletion portends poor prognosis        |
|                                          | Oligodendroglioma, IDH-mutant and 1p/19q codeleted                     | Diffuse glioma with IDH1 or IDH2 mutation and whole-arm codeletion of chromosomes 1p and 19q                                                                                                   | IDH1/2, -1p/19q, TERT promoter, CIC, NOTCH1, FUBP1                                                                                                                   | 2–3, depending on presence of anaplasia, necrosis, microvascular proliferation, and mitoses | Resection and radiation ± PCV therapy. Indolent course, similar to IDH-mutant gliomas                                                           |
| <b>Pediatric-type low grade gliomas</b>  | Diffuse astrocytoma, MYB- or MYBL-altered                              | Diffuse, low grade glioma with MYB or MYBL alterations                                                                                                                                         | MYB or MYBL fusions involving PCDHGA1, MAML2, or MMP16                                                                                                               | 1                                                                                           | Prognosis generally good after resection                                                                                                        |
|                                          | Angiocentric glioma                                                    | Diffuse, low-grade glioma with MYB alteration and angiocentric growth                                                                                                                          | MYB fusions (usually with QKI)                                                                                                                                       | 1                                                                                           | Prognosis generally good after resection                                                                                                        |
|                                          | Polymorphous low-grade neuroepithelial tumor of the young (PLNTY)      | CD34-positive neoplasm with infiltrative growth and MAPK pathway alterations                                                                                                                   | BRAF V600E, FGFR2/3, NTRK                                                                                                                                            | 1                                                                                           | Generally good after resection, but some data supports targeted therapy                                                                         |
|                                          | Diffuse low-grade glioma, MAPK pathway-altered                         | Infiltrative, low-grade glioma with mixed oligodendroglial and astrocytic morphology with MAPK pathway alteration and without IDH or histone mutations or CDKN2A deletion                      | BRAF V600E, FGFR1                                                                                                                                                    | 1                                                                                           | Generally good after resection, but some data supports targeted therapy                                                                         |
| <b>Pediatric-type high-grade gliomas</b> | Diffuse midline glioma, H3K27-altered                                  | Infiltrative, high-grade glioma arising in midline structures (usually pons but sometimes bithalamic) with characteristic H3K27 mutation                                                       | H3K27, TP53, EZHIP, EGFR, PDGFRA                                                                                                                                     | 4                                                                                           | Generally poor prognosis. Radiation and alkylating chemotherapy are standard                                                                    |
|                                          | Diffuse hemispheric glioma, H3G34-mutant                               | Infiltrative, high-grade glioma arising in cerebral hemisphere with H3 G34 mutant                                                                                                              | H3G34, TP53, ATRX                                                                                                                                                    | 4                                                                                           | Generally poor prognosis. Resection, radiation and alkylating chemotherapy are standard                                                         |
|                                          | Diffuse pediatric-type high-grade glioma, H3-wildtype and IDH-wildtype | Infiltrative, high-grade glioma, usually hemispheric, lacking in histone or IDH mutations                                                                                                      | PDGFRA (RTK1), EGFR/TERT promoter (RTK2), MYCN amplification (pHGG MYCN)                                                                                             | 4                                                                                           | Standard therapy similar to above. Poor prognosis even when MGMT promoter methylated                                                            |
|                                          | Infant-type hemispheric glioma                                         | Infiltrative, high-grade glioma arising in cerebral hemisphere with sole activating mutation in RTK                                                                                            | NTRK family, ROS, ALK, MET                                                                                                                                           | 4                                                                                           | Better prognosis than other pHGG due to effectiveness of targeted therapy                                                                       |

Table 1 (continued)

|                                         | Tumor type                                                    | Definition                                                                                                                                                                                             | Recurrent genetic abnormalities                                                | Grade                                                              | Management and prognosis                                                                                                              |
|-----------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Circumscribed astrocytic gliomas</b> | Pilocytic astrocytoma                                         | Circumscribed, usually posterior fossa proliferation of polarized, piloid cells ± eosinophilic granular bodies and Rosenthal fibers with MAPK pathway alteration                                       | KIAA1549-BRAF fusion, NF1, BRAF V600E                                          | 1                                                                  | Good with complete resection                                                                                                          |
|                                         | Subependymal Giant Cell Astrocytoma                           | Periventricular proliferation of large, ganglion-like astrocytes, usually associated with tuberous sclerosis                                                                                           | TSC1, TSC2                                                                     | 1                                                                  | MTOR inhibitors, resection                                                                                                            |
|                                         | Pleomorphic xanthoastrocytoma                                 | Usually hemispheric, circumscribed tumor with pleomorphic astrocytes and frequently eosinophilic granular bodies                                                                                       | BRAF V600E, CDKN2A/B                                                           | 2–3                                                                | Safe maximal resection. Recurrence and progression not uncommon                                                                       |
|                                         | Chordoid glioma                                               | Proliferation of epithelioid, GFAP-positive cells in floor of third ventricle with recurrent PRKCA mutation                                                                                            | PRKCA D463H                                                                    | 2–3                                                                | Maximal safe resection ± radiation                                                                                                    |
|                                         | Astroblastoma, MN1-altered                                    | Astrocytic proliferation with angiocentric growth and recurrent MN1 alteration                                                                                                                         | MN1                                                                            | Not established                                                    | Resection ± chemo/radiation therapy. Local recurrence common, but overall survival is good                                            |
| <b>Ependymal tumors</b>                 | High-grade astrocytoma with piloid features                   | Most common in posterior fossa. High-grade astrocytoma with MAPK alterations, distinct methylation profile, and morphology reminiscent of pleomorphic xanthoastrocytoma or pilocytic astrocytoma       | CDKN2A/B, NF1, BRAF, ATRX                                                      | Not definitively established, but most consistent with WHO grade 3 | Resection ± chemo/radiation therapy. Prognosis intermediate between grade 3 and grade 4 IDH-mutant astrocytoma                        |
|                                         | Supratentorial ependymoma, ZFTA fusion-positive               | Circumscribed glioma with variable appearance but at least focal perivascular or ependymal rosettes and ZFTA fusion                                                                                    | ZFTA fusions, most commonly with RELA. CDKN2A/B alterations frequently present | 2–3                                                                | Aggressive, especially when CDKN2A/B deletion is present                                                                              |
|                                         | Supratentorial ependymoma, YAP1 fusion-positive               | Circumscribed glioma with ependymal and perivascular rosettes and YAP1 fusions                                                                                                                         | YAP1 fusions, most commonly with MAMLD1                                        | 2–3                                                                | More indolent than ZFTA fusion-positive ependymomas. Good prognosis with resection                                                    |
|                                         | Posterior fossa group A (PFA) ependymoma                      | Posterior fossa location, PFA methylation profile or H3K27Me3 loss                                                                                                                                     | EZH2 overexpression, H3K27Me3 loss                                             | 2–3                                                                | Good with complete resection. 1q gain and incomplete resection indicate poor prognosis                                                |
|                                         | Posterior fossa group B (PFB) ependymoma<br>Spinal ependymoma | Posterior fossa location, PFB methylation profile, retained H3K27Me3<br>Intramedullary, circumscribed glioma in spinal cord with ependymal rosettes in fibrillary matrix. Distinct methylation profile | Monosomy 6, trisomy 18, chromosome 22q loss<br>NF2, MYCN amplification         | 2–3<br>2–3                                                         | Incomplete resection and 13q loss indicate poor prognosis<br>Prognosis good after resection but poor if MYCN amplification is present |

Table 1 (continued)

| Tumor type                                       | Definition                                                                                                                                     | Recurrent genetic abnormalities                     | Grade | Management and prognosis                                                                     |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-------|----------------------------------------------------------------------------------------------|
| Myxopapillary ependymoma                         | Ependymal tumor usually located at conus medullaris with angiocentric rosettes with myxoid change and microcysts                               | chromosomes +6/-10                                  | 2     | Generally good after complete resection in adults. CSF dissemination more common in children |
| Subependymoma                                    | Bland, nested ependymal proliferation in fibrillary matrix. Most common in 4th ventricle, usually does not enhance                             | Loss of 19, partial 6                               | 1     | Good prognosis with rare progression even after subtotal resection                           |
| Medulloblastoma, WNT-activated                   | Embryonal tumor arising in brainstem with WNT-activating mutation                                                                              | CTNNB1 and monosomy 6, APC                          | 4     | Excellent with therapy. Reduced radiation protocols can be considered                        |
| Medulloblastoma, SHH-activated and TP53-wildtype | Embryonal tumor arising in brainstem with SHH-activating mutation and without TP53 mutation                                                    | PTCH1, SUFU, SMO, MYCN                              | 4     | Intermediate, poor if MYCN amplification present                                             |
| Medulloblastoma, SHH-activated and TP53-mutant   | Embryonal tumor arising in brainstem with SHH-activating mutation and with TP53 mutation                                                       | PTCH1, SUFU, SMO, TP53, MYCN                        | 4     | Poor prognosis                                                                               |
| Medulloblastoma, non-WNT/non-SHH                 | Embryonal tumor arising in brainstem without SHH- or WNT-activating mutations                                                                  | Separated into 8 subgroups by methylation profiling | 4     | Groups 6 and 7 have best prognosis. Groups 2, 3, and 8 have the worst                        |
| Atypical Teratoid/Rhabdoid Tumor                 | Poorly differentiated tumor with rhabdoid morphology occurring anywhere throughout the neuraxis with characteristic loss of SMARCB1 or SMARCA4 | SMARCB1/INI1, SMARCA4                               | 4     | Poor prognosis                                                                               |
| Cribriform Neuroepithelial Tumor                 | Periventricular tumor with SMARCB1 loss and cribriform rather than rhabdoid morphology                                                         | SMARCB1/INI1                                        | 4     | Better response to therapy than AT/RT and good overall survival                              |
| Embryonal tumor with multilayered rosettes       | Mostly intracranial tumor with primitive cells arranged into rosettes and either C19MC or DICER1 mutation                                      | C19MC, DICER1                                       | 4     | Aggressive course with survival times around 1 year even with intensive therapy              |
| CNS neuroblastoma, FOXR2-activated               | Primitive neuroepithelial tumor with FOXR2 activating mutation                                                                                 | FOXR2                                               | 4     | New entity with limited survival data                                                        |
| CNS tumor with BCOR internal tandem duplications | Primitive neuroepithelial tumor with abundant pseudorosettes and internal tandem duplications in exon 15 of BCOR                               | BCOR                                                | 4     | New entity with limited survival data                                                        |

behavior or prognosis as a grade 2 tumor in WHO classifications of other neoplasms elsewhere in the body, the official usage is “CNS WHO grade 2” not simply “WHO grade 2.” However, for ease of reading, the latter approach is adopted in this review.

Other changes in general nomenclature include “not otherwise specified (NOS)” and “not elsewhere classified (NEC)” [5]. NOS means that the molecular testing required to classify a CNS lesion is not available. For example, if a supratentorial lesion has ependymal morphology, but sequencing and methylation profiling are not available, a final diagnosis of “Supratentorial ependymoma, NOS” would be appropriate. NEC, on the other hand, means that the appropriate molecular testing was performed but did not provide enough useful information for further classification. Thus, if molecular testing was performed on that supratentorial ependymoma but failed to uncover a *ZFTA* or *YAP1* fusion, it would be called “Supratentorial ependymoma, NEC.”

## Adult-Type Diffuse Gliomas

Diffuse gliomas, accounting for ~70% of adult brain tumors, are the most common type of primary brain tumor to arise in adults [6] (the most common CNS neoplasm overall is metastatic disease). Prior to 2016, morphologic features drove the classification of all diffuse gliomas. Tumors with round, uniform nuclei and cytoplasmic clearing were referred to as oligodendrogliomas, those with clumped chromatin and angulated nuclei were referred to as astrocytomas, and those with intermediate features were called oligoastrocytomas [7]. In the 2016 edition, molecular features were introduced into the classification of gliomas, with *IDH* mutation and 1p/19q codeletion required for the diagnosis of oligodendroglioma, and *IDH* mutation status and histologic grade used to parse astrocytomas into subtype [1]. In the 2021 classification, diffuse gliomas are now sorted into three basic types by morphology and molecular features with, grading done within each type. Hybrid entities like oligoastrocytoma, which are nearly always classified as other entities when molecular testing is performed, are no longer listed [8, 9].

One key difference between the 2016 and 2021 WHO classifications is in the way that *IDH*-wildtype gliomas are defined and graded. Although high-grade morphologic features such as mitoses, necrosis, and microvascular proliferation are still considered, all tumors lacking *IDH* mutations that have concomitant gain of chromosome 7 and loss of chromosome 10, *EGFR* amplification, or *TERT* promoter mutations are called glioblastoma and are given a WHO grade of 4 [10]. These tumors, which tend to occur in older adults and are rare below the age of 55, are highly aggressive, with death occurring within 15–18 months for most

patients even with chemotherapy and radiation [11]. For most patients, symptoms related to mass effect develop rapidly, and high-grade imaging features, such as peripheral enhancement and central necrosis, are usually present at diagnosis. The current standard treatments include maximal surgical resection (when anatomically feasible), radiation, and temozolomide [12–14]. *IDH* wildtype glioblastoma is a morphologically, genetically heterogeneous category comprised of multiple different subtypes, including a small cell type (which mimics oligodendroglioma), a granular cell type with PAS-positive cytoplasmic inclusions, an epithelioid type with well-defined cytoplasmic borders and ample, eosinophilic cytoplasm, a giant cell type, and a sarcomatous type that may lose GFAP and olig2 expression and contain heterologous elements. By definition, *IDH*-wildtype glioblastomas are negative for *IDH1* R132H, and the majority express markers of glial differentiation and retain normal *ATRX* by immunohistochemistry (meaning that no *ATRX* mutation is present). In addition to the aforementioned *EGFR*, *TERT* promoter, and +7/-10 phenotypes in the diagnostic criteria, other common molecular abnormalities include *CDKN2A/B* deletion, *PTEN* alterations, *TP53* mutations, *MDM2* or *MDM4* amplification, *BRAF* V600E mutations (especially the epithelioid subtype), and *MGMT* promoter methylation [15–17]. Of these, *MGMT* promoter methylation status is the most critical, as it predicts response to alkylating chemotherapeutic drugs such as temozolomide and lomustine [13, 18, 19]. *MGMT* promoter methylation status predicts both overall and progression-free survival in elderly patients with glioblastoma treated with alkylating agents in addition to radiotherapy [20]. *PTEN* status may also be important for predicting response to therapy. Some studies show that *PTEN* mutations, which are present in about 40% of gliomas, may render them more sensitive to radiation therapy [21]. One study suggested that, in patients with *EGFR* amplification, targeted *EGFR* inhibitors may only be effective in patients with intact *PTEN* expression [22]. A handful of case reports also suggest that *BRAF* inhibitors may be effective only when *BRAF* V600E mutations are present [23, 24]. More trials are needed in order to determine which groups of patients might benefit from targeted therapies.

Astrocytoma, *IDH*-mutant, is defined by a change-of-function mutation in *IDH1* or *IDH2* resulting in overproduction of the oncometabolite D-2-hydroxyglutarate, which acts as an inhibitor of enzymes that use  $\alpha$ -ketoglutarate as a cofactor, such as certain DNA demethylases, resulting in genomic CpG hypermethylation and suppression of differentiation [25]. They most often occur in younger adults (median age 38) and are rarely diagnosed in adults over the age of 55. Patients usually present with seizures and are found to have diffuse, T2 FLAIR hyperintense, supratentorial masses with little or no enhancement [26]. The majority of *IDH*-mutant astrocytomas also have *TP53*

alterations resulting in strong nuclear accumulation of abnormal p53 in >50% of tumor cell nuclei. About 90% of supratentorial *IDH*-mutant astrocytomas also have *ATRX* mutations that result in loss of normal *ATRX* expression in the tumor cells [27, 28]. This makes p53 and *ATRX* immunostains useful as part of a panel when working up diffuse gliomas—a young patient with a glioma that displays strong p53 expression and loss of *ATRX* is likely to have a non-canonical *IDH* mutation if *IDH1* R132H immunostain is negative [27]. *IDH*-mutant astrocytoma grades range from 2 to 4, based on the presence of anaplasia, mitotic activity, necrosis, microvascular proliferation, and homozygous *CDKN2A/B* deletion [29]. One major change between the 2016 and 2021 WHO Classifications is that, even when enough high-grade features are present to warrant a grade 4 designation, *IDH*-mutant astrocytomas are no longer referred to as glioblastomas, since even high-grade *IDH*-mutant astrocytomas are less aggressive than their *IDH*-wildtype counterparts [30]. Nevertheless, the current recommended standard of therapy for high-grade *IDH*-mutant gliomas remains similar to that for *IDH*-wildtype glioblastoma [12], although this may change as clinical trial target patients based on *IDH* status as well as tumor grade.

In addition to whole arm 1p/19q codeletion, *IDH* mutations are required for the diagnosis of oligodendroglioma, reflecting the fact that true 1p/19q codeletion occurring from unbalanced translocation is always seen in conjunction with *IDH1* or *IDH2* mutations [29, 31–33]. Like *IDH*-mutant astrocytomas, these are relatively less aggressive tumors that primarily occur in younger adults (median age 43), although gliomatosis cerebri-like growth patterns and seeding of the cerebrospinal fluid can sometimes be seen in more advanced stages. In addition to *IDH* mutations and 1p/19q codeletion, the majority of oligodendrogliomas also have *TERT* promoter mutations [32]. In contrast to *IDH*-mutant astrocytomas, oligodendrogliomas tend to have retained *ATRX* expression and lack accumulation of p53, since 1p/19q co-deletion is essentially mutually exclusive with *TP53* and *ATRX* alterations [34, 35]. Some oligodendrogliomas have *CDKN2A/B* deletion, which predicts more aggressive behavior when present [36]. As was the case in the 2016 edition, grades range from 2 to 3 based on the presence of anaplasia, mitotic activity, necrosis, and microvascular proliferation [2]. Currently, radiation followed by adjuvant procarbazine, lomustine, and vincristine (PCV therapy) is the recommended treatment protocol for oligodendrogliomas [37, 38].

## Pediatric-Type Diffuse Gliomas

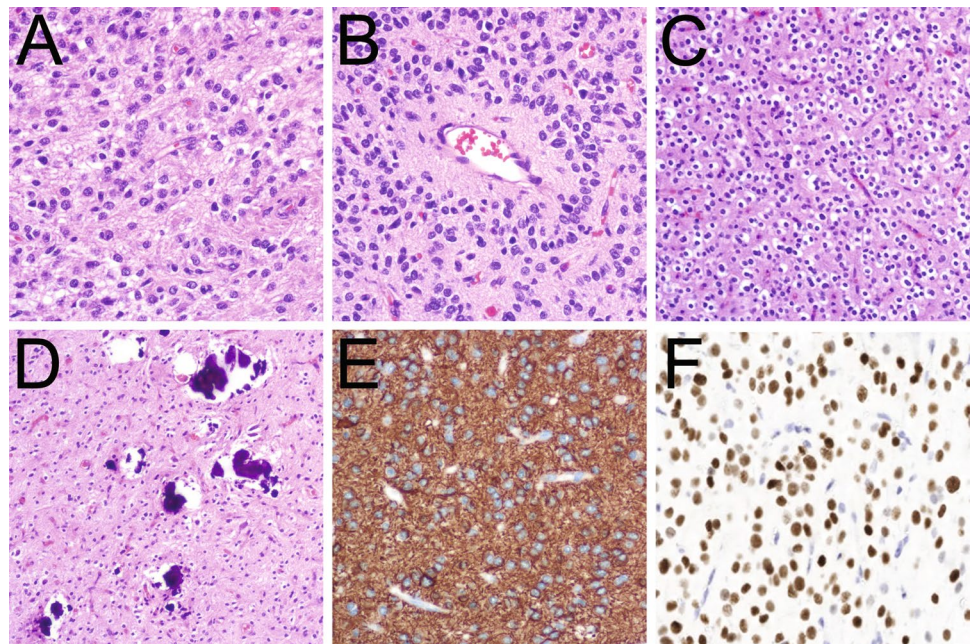
Since the 2016 WHO Classification was published, understanding of the distinct biology of pediatric-type diffuse gliomas has exploded, resulting in the addition of distinct chapters containing eight newly added tumor types [2].

Four are classified as diffuse low-grade gliomas, reflecting their relatively indolent clinical behavior despite lack of a clear tumor: nontumor border. All are characterized by mutations that result in mitogen-activated protein kinase (MAPK) pathway activation and by the absence of *IDH* or histone mutations [39]. The first of these, “diffuse astrocytoma, *MYB*- or *MYBL1*-altered,” is defined by *MYB*- or *MYBL1*- fusions, with the most common fusion partners being *PCDHGA1*, *MMP16*, and *MAML2* [40–43]. Angiocentric gliomas are only distinguishable from diffuse astrocytoma, *MYB*- or *MYBL1*-altered by clustering of tumor cells around blood vessels, and a different *MYB* fusion partner (usually *QKI*) [40–42, 44]. “Polymorphous low-grade neuroepithelial tumor of the young (PLNTY)” typically has a mixture of cells with oligodendroglial and astrocytic morphology with aberrant CD34 expression (Fig. 1), and frequently features perivascular rosettes and coarse calcifications. PLNTYs have a variety of MAPK-activating alterations, including *BRAF* V600E mutations, *NTRK* alterations, and fusions involving *FGFR2* or *FGFR3* [45]. “Diffuse low-grade glioma, MAPK pathway-altered” commonly has *BRAF* V600E mutations or *FGFR1* alterations, much like both PLNTY and low-grade glioneuronal tumors (e.g., dysembryoplastic neuroepithelial tumor and ganglioglioma) [42, 46]. All four of these tumor types occur in the cerebral hemispheres, have low-grade features on imaging, often present with seizures, and show a predilection for teenagers or young adults [40, 42, 44]. All four also show closely related DNA methylation profiles [47], so it remains to be seen whether they will ultimately remain distinct tumor types in future editions of the WHO classification. All have targetable MAPK pathway alterations, and early studies using targeted therapies have been promising [48–50].

Four of the pediatric-type diffuse gliomas are classified as high-grade: “diffuse midline glioma, H3 K27-altered;” “diffuse hemispheric glioma, H3 G34-mutant;” “diffuse pediatric-type high-grade glioma, H3-wildtype and *IDH*-wildtype;” “infant-type hemispheric glioma” [2, 26, 51]. The latter three were newly added to the 2021 edition of the WHO Classification. Like the low-grade pediatric gliomas, all four lack *IDH* mutations and should be considered in young adult patients with *IDH* wildtype gliomas. Both diffuse midline glioma, H3 K27-altered and diffuse hemispheric glioma, H3 G34-mutant are driven by histone mutations. For reasons that are not entirely understood, tumors with H3 K27 mutations tend to occur in midline structures, while those with H3 G34 mutations tend to occur in the cerebral hemispheres [52–56]. Although diffuse midline gliomas have somewhat variable morphology, with some having high proliferation rates and necrosis like the high-grade gliomas seen in adults (Fig. 2) and others lacking those features, they all have decreased H3K27Me3 by immunohistochemistry. In diffuse midline gliomas, such



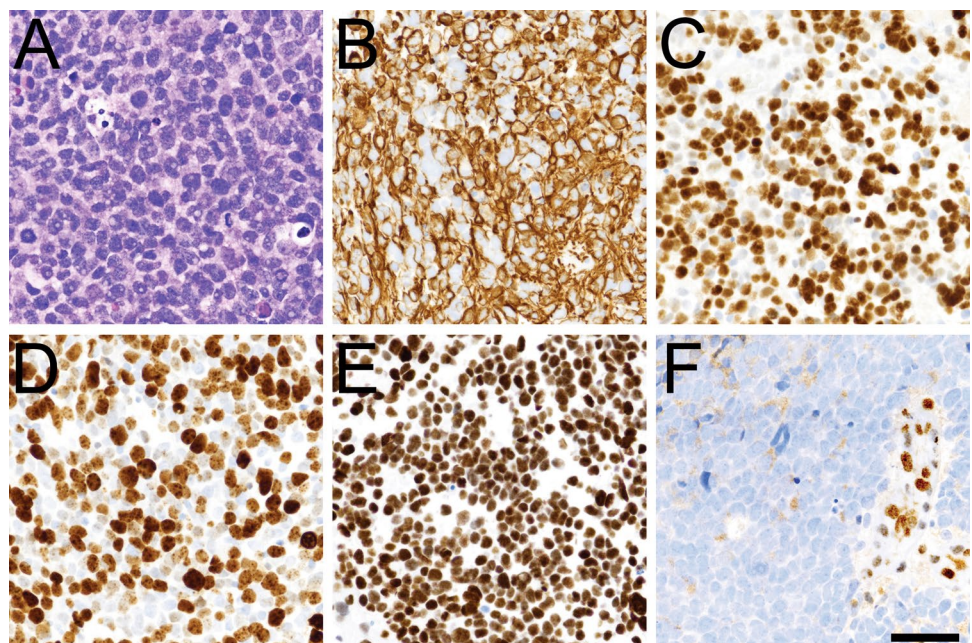
**Fig. 1** Polymorphous low-grade neuroepithelial tumor of the young. In keeping with the “polymorphous” descriptor, polymorphous low-grade neuroepithelial tumor of the young can have a variety of appearances, including that of a diffuse glioma (A), ependymoma (B), and oligodendroglioma (C). These tumors can have abundant mineralization (D) and show abundant CD34 positivity (E) and OLIG2 nuclear staining (F). This particular tumor had a BRAF V600E mutation, and clustered among PLNTYs by DNA methylation profiling



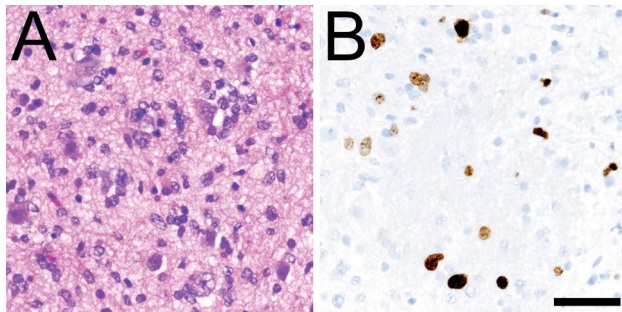
loss occurs by a lysine-to-methionine substitution in the histone H3 protein, resulting in inhibition of the EZH2 catalytic subunit of the polycomb repressive complex 2 (PRC2) protein [57]. In histone-mutated hemispheric gliomas, it occurs when substitution of a glycine at position 35 for arginine or valine results in reduced binding of SETD2 and KDM2A to the tail of the histone H3 protein, resulting in diminished H3 K37 Me3 [58, 59]. Both mechanisms lead to increased proliferation and decreased differentiation, and both histone-mutant gliomas are aggressive tumors with uniformly poor prognoses [51, 52, 60]. The third tumor, high-grade

glioma, H3-wildtype and IDH-wildtype, is a category that encompasses high-grade diffuse gliomas that lack both histone and *IDH* mutations and can have multiple driver mutations (Fig. 3). Some have similar driver mutations as adult IDH-wildtype gliomas, including *EGFR*, *PDGFRA*, *TP53*, and *NF1*, but their methylation profiles are distinct from adult IDH-wildtype gliomas [61]. Three different subgroups have been identified by DNA methylation profiling: pHGG RTK1, pHGG RTK2, and pHGG *MYCN*. Tumors of the pHGG RTK1 subtype most frequently have *PDGFRA* alterations and are the type most frequently found in patients

**Fig. 2** Diffuse midline glioma, H3 K27-altered. Diffuse midline gliomas tend to look like most other diffusely infiltrative gliomas (A), including immunopositivity for GFAP (B) and OLIG2 (C). In keeping with their high-grade nature, Ki67 is usually quite high (D). H3 K27M-specific antibody often shows robust nuclear staining (E); those same tumor cells will be weak to negative for H3K27me3, whereas admixed nonneoplastic cells will still be positive (F). Scale bar = 50 microns in all panels





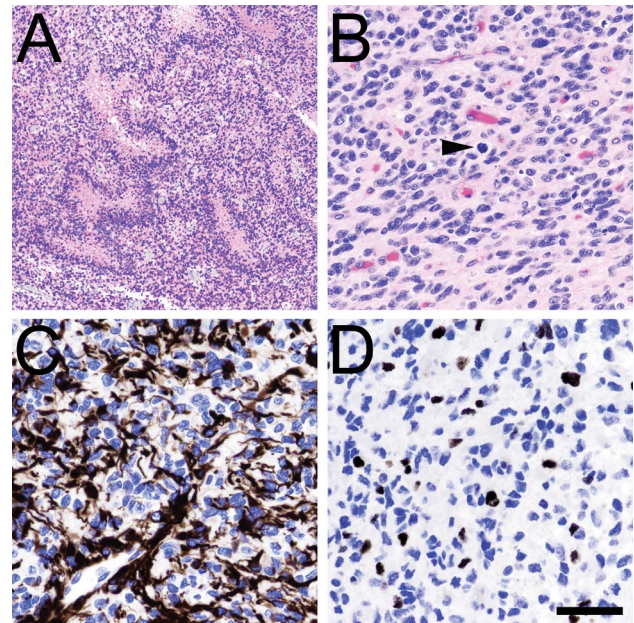


**Fig. 3** Diffuse pediatric-type high-grade glioma, H3-wildtype and IDH-wildtype. This diffuse pediatric-type high-grade glioma, H3 wildtype and IDH-wildtype, looked like a typical diffuse glioma infiltrating into surrounding brain tissue (**A**); many tumor cells will be Ki67 positive (**B**). This particular tumor had *MYCN* amplification and *TP53* mutation, mapped to diffuse pediatric-type high-grade glioma, H3 wildtype, and IDH-wildtype by DNA methylation profiling, and had already disseminated throughout the cerebrospinal fluid at the time of clinical presentation. Scale bar = 50 microns in both panels

with Lynch syndrome. pHGG RTK2 tumors most frequently have *EGFR* amplification and *TERT* promoter mutations, while pHGG *MYCN* tumors have *MYCN* activation (usually amplification) [61]. Like pediatric high-grade gliomas with histone mutations, these tumors are highly aggressive with poor prognoses, with 2-year survival rate of 23.5% and median overall survival of only 17 months, even when *MGMT* promoter methylation is present [54]. Infant-type hemispheric glioma (Fig. 4) is a large, hemispheric mass that typically occurs during the first year of life and is usually driven by RTK-activating fusions, including those in the *NTRK* family, *ROS1*, *ALK*, or *MET* [62]. Despite their histopathologic similarity to IDH-wildtype glioblastomas, these have a better prognosis than the other three pediatric-type high-grade gliomas, with 5-year survival rates ranging from 25 to 50%, and there are some data suggesting that they respond to drugs targeting whichever RTK is altered [63].

### Circumscribed Astrocytic Gliomas

Gliomas with more well-delineated borders separating them from the surrounding brain parenchyma, previously referred to as “other astrocytic tumors,” are now categorized as circumscribed astrocytic gliomas. This category includes pilocytic astrocytoma, subependymal giant cell tumor, pleomorphic xanthoastrocytoma, chordoid glioma (Table 1), “astroblastoma, *MNI*-altered,” and “high-grade astrocytoma with piloid features” [2, 51]. Most of the entities in this category are well established with only minor changes in the names, e.g., astroblastoma, *MNI*-altered and chordoid glioma. Astroblastomas, with their perivascular rosettes and reverse nuclear polarity, share some



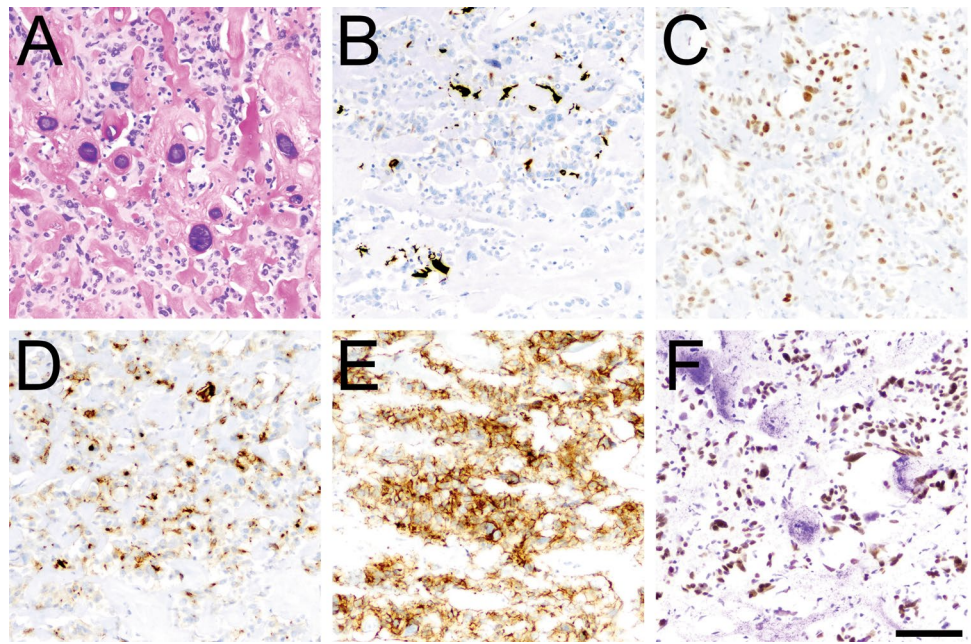
**Fig. 4** Infant-type hemispheric glioma. Morphologically, infant-type hemispheric gliomas may be indistinguishable from adult-type IDH<sup>wt</sup> glioblastomas, with palisading necrosis (**A**), abundant mitoses (**B**), variable GFAP positivity (**C**), and elevated Ki67 (**D**). However, they will not have the same molecular profile as glioblastomas; this tumor had an isolated *NTRK* fusion, and mapped to infant-type hemispheric glioma by DNA methylation profiling. Scale bar = 250 microns in **A**, 50 microns in **B–D**

histomorphologic features with ependymal and embryonal tumors (Fig. 5) but are now defined by *MNI* alterations [64–66]. Patients generally do well following surgical resection, but when anatomy precludes complete excision, chemotherapy and radiation offer some benefit [67]. Chordoid glioma (formerly known as “chordoid glioma of the third ventricle”), which nearly always occurs in the anterior third ventricle of adults, is comprised of spindled to epithelioid TTF1- and GFAP-positive cells in a myxoid stroma, a recurrent p.D463H missense mutation in the *PRKCA* gene, and is frequently separated from the adjacent brain parenchyma with a dense, lymphoplasmacytic infiltrate [68, 69].

High-grade astrocytoma with piloid features, the only new circumscribed astrocytic glioma, is an aggressive astrocytic neoplasm with a combination of MAPK pathway-activating alterations (e.g., involving *NF1*, *FGFR*, or *BRAF*), *ATRX* mutations (manifesting as loss of normal ATRX immunostain), and homozygous *CDKN2A/B* deletion [70]. It can occur anywhere in the central nervous system but most often arises in the posterior fossa, and typically occurs in middle-aged adults [70]. The morphologic features are variable and can resemble glioblastoma or pleomorphic xanthoastrocytoma (PXA); features reminiscent of pilocytic astrocytoma, such as eosinophilic granular bodies, Rosenthal fibers, and long, delicate processes are only seen in a third of cases.



**Fig. 5** Astroblastoma. Astroblastomas can show a variety of morphologies, including heavy sclerosis and mineralization (A). Tumor cells show variable GFAP immunopositivity (B) but are generally positive for OLIG2 (C), epithelial membrane antigen (D), podoplanin (E), and SATB2 (F). Scale bar in F = 100 microns in all panels



While most of these tumors arise *de novo*, a small subset arise in pre-existing grade 1 pilocytic astrocytoma and were previously called “anaplastic pilocytic astrocytomas” [70]. Recent studies using genomic DNA methylation profiling revealed that high-grade astrocytoma with piloid features have their own distinct cluster [70]. There is only one retrospective study with any data on long-term prognosis, but the 5-year survival rate is roughly 50% and overall survival appears to be similar to that for patients with astrocytoma, IDH-mutant, WHO grade 4 [70]. As with some of the other more recently defined entities, it remains to be seen whether high-grade astrocytomas with piloid features respond better to conventional therapies versus targeted therapies.

Sometimes, it can be difficult to distinguish between circumscribed and diffuse astrocytic gliomas. PXAs, for example, can look very similar to epithelioid glioblastomas, especially if the PXA is grade 3 with numerous mitoses and/or necrosis. Both are *IDH* wildtype, both can have *CDKN2A/B* deletion and MAPK-activating alterations (e.g., BRAF V600E), and both can arise in similar locations and age groups. Genomic methylation profiling is particularly helpful in such circumstances, as many tumors thought to be epithelioid glioblastomas actually map either to PXA or pediatric high-grade glioma, RTK1 subtype [71].

## Ependymal Tumors

Because of advanced molecular testing techniques such as DNA methylation profiling, categorization of ependymomas has completely changed from being based on morphology to being based on location and molecular alteration [72].

Tanycytic, clear cell, and papillary are listed as morphologic phenotypes rather than separate subtypes, and anaplastic ependymoma is no longer listed as an entity [2]. The majority of supratentorial ependymomas have fusions involving either *ZFTA* or *YAP1* and are derived from radial glial cells [73]. *ZFTA* fusion-positive ependymomas are more common, comprising 25–58% of all supratentorial ependymomas in adults and 66–84% in children [74–76]. The most common fusion partner is *RELA*, and homozygous deletion of *CKDN2A/B*, which predicts more aggressive clinical behavior, may be present [77]. *YAP1* fusion-positive ependymomas, which show more indolent behavior than *ZFTA* fusion-positive ependymomas, are rare and tend to occur primarily in young children [78]. Other genetic alterations predicting their clinical behavior have not yet been well studied. Either can be assigned a grade of 2 or 3, depending on the presence of mitotic activity and microvascular proliferation [79].

Posterior fossa ependymomas are sorted into PFA and PFB categories by DNA methylation profiling [80]. Like diffuse midline gliomas, PFA ependymomas show loss of H3 K27 trimethylation; however, only a small subset (~4%) have histone mutations [81]. Instead, PFA ependymomas express more Enhancer of Zest Homologs Inhibitory Protein (EZHIP), which binds to and inhibits the catalytic domain of PRC2 in a similar manner to the H3 K27M mutated histone protein [81, 82]. Such EZHIP overexpression also drives the rare diffuse midline gliomas that lack histone mutations, and in both PFA ependymomas and diffuse midline gliomas, H3 histone mutations and EZHIP alterations are mutually exclusive [83]. PFB ependymomas usually show whole chromosome abnormalities such as 22q loss, monosomy 6, and

monosomy 18 [80, 84]. In addition, PFB ependymomas are more likely to occur in older adults and adolescents and are less aggressive than PFA ependymomas. Gain of chromosome 1q and incomplete resection are poor prognostic indicators for all types of posterior fossa ependymomas, and loss of 13q suggests worse behavior in PFB ependymomas [85].

Spinal ependymomas, which are their own cluster on DNA methylation profiling, have 22q loss like PFB ependymomas [84]. These occur more frequently as intramedullary masses in the cervicothoracic region of the spinal cord, in contrast to myxopapillary ependymoma, which occur in the lumbar region [86]. For the most part, spinal ependymomas have good outcomes and long progression free survival times, except for the subset with *MYCN* amplification, which spread throughout the CSF [87]. Myxopapillary ependymoma has its own distinct methylation cluster and was historically given a WHO grade of 1 due to its generally good prognosis and bland appearance [1]. But, because of the possibility of CSF dissemination, which is more common in younger patients, and because so many patients live with persistent disease requiring multiple operations and radiotherapy, they are now considered WHO grade 2 in the 2021 classification [88–90].

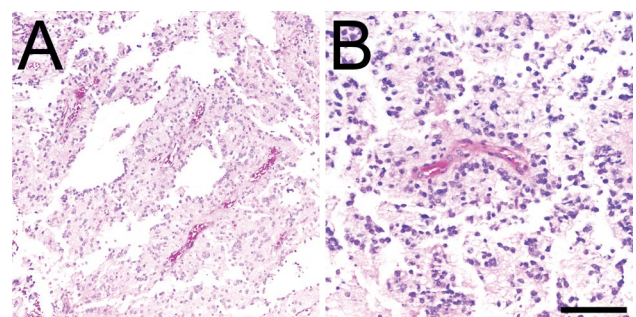
Subependymomas, which are still considered WHO grade 1, are indolently growing glial tumors with subependymal morphology. Approximately two-thirds of them arise in the fourth ventricle, with the remainder being supratentorial [91]. Like ependymomas, they can show cystic changes and calcification on imaging but are less likely to show much enhancement after contrast administration [92]. The most common genetic alterations are the loss of chromosome 19, partial loss of chromosome 6, and even H3 K27 mutations, all three of which can occur in brainstem subependymomas [84]. Overall, the clinical behavior of these tumors is benign with occurrence being rare even after subtotal resection, even when H3 K27 mutations are present [91, 92].

## Embryonal Tumors

Like pediatric gliomas and ependymal tumors, DNA methylation profiling has resulted in an explosion in the number of embryonal tumor types, both from the splitting of existing tumor types into additional categories and from the addition of newly characterized types of tumors [93, 94]. Medulloblastomas, the most common embryonal tumors, are divided into four different molecular subtypes, including *WNT*-activated, *SHH*-activated and *TP53*-wildtype, *SHH*-activated and *TP53*-mutant, and non-*WNT*/non-*SHH*. Pediatric *WNT*-activated medulloblastomas respond extremely well to existing therapies and have excellent prognoses, but retain a WHO grade of 4 to reflect their aggressive nature without treatment [93, 95–97]. *SHH*-activated tumors are

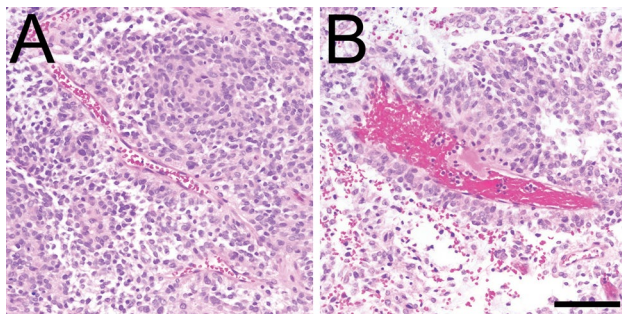
more aggressive if *TP53* mutations or *MYCN* amplification are present [96]. Non-*WNT*/non-*SHH* tumors, formerly known as group 3 and group 4 medulloblastomas, comprise the majority of medulloblastomas. As is the case with *SHH*-activated tumors, *MYCN* amplification is a poor prognostic indicator [98]. Non-*WNT*/Non-*SHH* tumors can be further divided into 8 different categories by DNA methylation profiling, with subgroups 2, 3, and 8 having the worst outcomes and groups 6 and 7, which are more likely to have favorable cytogenetic aberrations, such as gain of chromosome 7 and loss of chromosomes 8 or 11, having the best [93].

Three new entities have been added to embryonal tumors. The first of these, cribriform neuroepithelial tumor, is a provisional entry. It tends to occur in periventricular locations and has *SMARCB1* mutations and *INI1* loss, much like atypical teratoid rhabdoid tumors [99, 100]. Unlike atypical teratoid rhabdoid tumor, however, cribriform neuroepithelial tumors lack rhabdoid morphology and have favorable response to therapy with average survival times of approximately 10 years [99]. CNS neuroblastoma, *FOXR2*-activated, consists of sheets of small, mitotically active, primitive cells with high N/C ratios that form occasional rosettes (Fig. 6). Like neuroblastomas, they can harbor areas of neuropil and ganglion cells. They are rare and newly classified, so there is currently little data on their prognosis [101]. CNS tumor with *BCOR* internal tandem duplication is defined by a heterozygous internal tandem duplication in exon 15 of *BCOR* [102]. They have small, uniform, ovoid to spindle-shaped cells, often arranged in perivascular pseudorosettes that can easily be confused with those in ependymal tumors or astroblastomas (Fig. 7), and can also have palisading necrosis reminiscent of that in glioblastomas. *BCOR* internal tandem duplications do not co-occur with *ZFTA* or *YAPI* fusions,



**Fig. 6** CNS neuroblastoma, *FOXR2*-activated. CNS neuroblastoma with *FOXR2*-activation features sheets of tumor cells with small round nuclei and a high nuclear:cytoplasmic ratio like most other embryonal tumors, although the tumor shown here had more abundant neuropil (A). Like many other new tumor entities, these tumor cells often arrange themselves in a perivascular pattern (B). Most such tumors will also show other high-grade features like necrosis and mitoses, but this one did not. Still, it clearly mapped to CNS neuroblastoma with *FOXR2*-activation by DNA methylation profiling. Scale bar = 250 microns in A, 100 microns in B





**Fig. 7** CNS tumor with *BCOR* internal tandem duplication. Many CNS tumors with *BCOR* internal tandem duplication will look like anaplastic ependymomas, including this one (**A**, **B**). But these tumors have a unique molecular profile, and their own DNA methylation pattern. Scale bar = 100 microns

*MNI* alterations, *IDH* mutations, or *H3* mutations, and DNA methylation profiling can be used to reliably distinguish them from other morphologically similar entities [102]. These tumors are rare, which limits prognostic data, but case series suggest the overall prognosis to be poor [102].

## Meningioma

Meningothelial tumors are all categorized as one heterogeneous type, with 15 distinct morphologic subtypes that range in WHO grade from 1 to 3 [2]. Numerous studies characterizing the molecular alterations seen in meningiomas, and how they correspond to location and morphology, have been performed [103–106]; however, meningiomas are still mostly graded the same way they were in the 2016 classification with two exceptions. Since *TERT* promoter mutations and homozygous deletion of *CDKN2A/B* have been associated with more aggressive clinical behavior, these are now sufficient for a WHO grade of 3 even if the histopathology does not otherwise meet the criteria [107, 108]. Complete resection is still the primary way meningiomas are managed; however, mutations known to lead to more aggressive behavior can be used to help guide the decision for adjuvant radiation or chemotherapy.

## Conclusions

The 2021 WHO CNS tumor classification draws on a wealth of data from molecular testing that led to the creation of new entities and more accurate stratification of pre-existing entities. Future clinical trials can recruit more homogeneous categories of patients, which will allow for better understanding of how tumor biology corresponds to treatment response and for the development of more targeted therapies. However, more multi-institutional and multi-national studies will

be needed to address the less common tumor types. More long-term prognostic studies will also be needed to better characterize some of the newer entities.

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s13311-022-01249-0>.

**Required Author Forms** Disclosure forms provided by the authors are available with the online version of this article.

**Funding** CH was supported by National Institute for Health R01NS102669, R01NS117104, R01NS118039, the Northwestern University P50CA221747 SPORE in Brain Tumor Research, and the Lou and Jean Malnati Brain Tumor Institute at Northwestern.

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